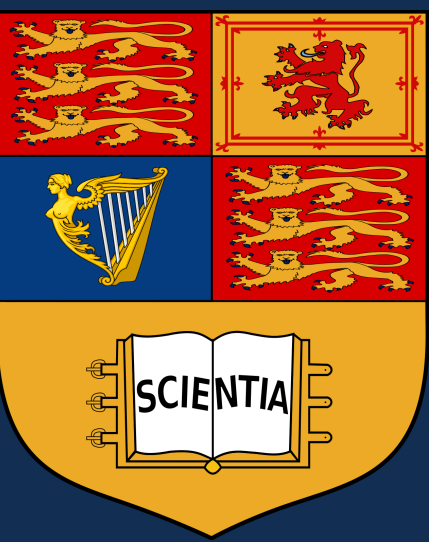




Introduction

In group theory, we classify groups by factoring them into products of smaller ones. One important notion of group products is the direct product. However, the direct product $G \times H$ imposes the condition that elements from G commutes with elements from H . We'll see that the condition disables us to factor groups such as D_6 . This poster introduces a useful generalization of direct product: the semidirect product.

- Motivation: Internal Product
- Building Larger Groups: Outer Product
- Examples and Visualization
- An Application: Groups of Order 30

Motivation: Internal Semidirect Product

The group $D_6 = \langle s, r | s^2 = r^3 = e, sr = r^{-1}s \rangle$ has subgroups $A = \{e, r, r^2\}, B = \{1, s\}$. One notes that $D_6 = AB$ and every element has a unique factorization ab for $a \in A, b \in B$. Thus, the map $\phi : (a, b) \mapsto ab$ from $A \times B$ to D_6 is a bijection. However it is not a group isomorphism because D_6 is non-abelian.

So what operation on $A \times B$ makes ϕ into a group isomorphism? We must require that for $a_1, a_2 \in A, b_1, b_2 \in B$:

$$\begin{aligned}\phi((a_1, b_1)(a_2, b_2)) &= \phi(a_1, b_1)\phi(a_2, b_2) \\ &= a_1b_1a_2b_2 \\ &= a_1(b_1a_2b_1^{-1})(b_1b_2) \\ &= a_3b_3\end{aligned}$$

where $a_3 = a_1(b_1a_2b_1^{-1})$, $b_3 = b_1b_2$ (by normality of A , $b_1a_2b_1^{-1} \in A$). This tells us the sensible operation $\circ$ on $A \times B$ is:

$$(a_1, b_1) \circ (a_2, b_2) = (a_1(b_1a_2b_1^{-1}), b_1b_2) \quad (1)$$

If we view $b_1a_2b_1^{-1}$ as b acting on a , (1) gives motivation for how to turn this construction around and build a group structure on $\{(a, b) | a \in A, b \in B\}$: you relate them by an action of B on A . (See below)

Definition: External Product

Definition: Let H and K be groups, $\phi : K \rightarrow \text{Aut}(H)$ a homomorphism. The group $H \rtimes_{\phi} K$ is the set $\{(h, k) | h \in H, k \in K\}$ with operation

$$(h_1, k_1)(h_2, k_2) = (h_1\phi_{k_1}(h_2), k_1k_2)$$

Where $\phi_{k_1}(h_2)$ is the automorphism $\phi(k_1)$ of H evaluated at h_2 . The product in each component on RHS is understood as a product in H or in K .

Examples

Notation: Let H, K be groups. When $H \rtimes_{\psi} K$ is understood, we write h for $(h, 1)$ and k for $(k, 1)$

- (Semidirect Generalizes Direct) Let H and K be any groups, $\phi : K \rightarrow \text{Aut}(H)$ the trivial homomorphism. Then

$$(h_1, k_1)(h_2, k_2) = (h_1\phi_{k_1}(h_2), k_1k_2) = (h_1h_2, k_1k_2)$$

$H \rtimes_{\phi} K \cong H \times K$ is just the direct product of H and K .

- (Dihedral Group) Let $H = \mathbb{Z}_n, K = C_2 = \langle k \rangle, \psi : K \rightarrow \text{Aut}(H)$ sending e to the identity map and k to inversion on H ($h \mapsto h^{-1}$). Externally $khk^{-1} = h^{-1}$, so $H \rtimes_{\psi} K \cong D_{2n}$

- (Creating Groups) $H = \mathbb{Z}_3, K = \mathbb{Z}_4 = \langle k \rangle, \psi : K \rightarrow \text{Aut}(H)$ sending

$$k^n \mapsto \begin{cases} \text{The identity map on } H, & \text{if } n \text{ is even} \\ \text{inversion on } H, & \text{if } n \text{ is odd} \end{cases}$$

$G := \mathbb{Z}_3 \rtimes_{\psi} \mathbb{Z}_4$ is not isomorphic to A_4 or D_{12} . Obviously G has an element of order 4 but neither A_4 or D_{12} does.

- (Non-Example) Fact: The Hamilton's quaternions group Q_8 IS NOT a semidirect product of its subgroups.

Most examples in this section comes from [2]

Interlude: Visualization of Semidirect Products

Cayley Diagram visualizes groups. Let G be a group generated by the set S . Every node represents an element g in G , each color represents an element s in S , and a colored directed edge between nodes indicates left multiplication of the source node by the color to get the target node. Figure 1 below give an example of visualizing $V_4 \rtimes_{\theta} C_3[1]$:

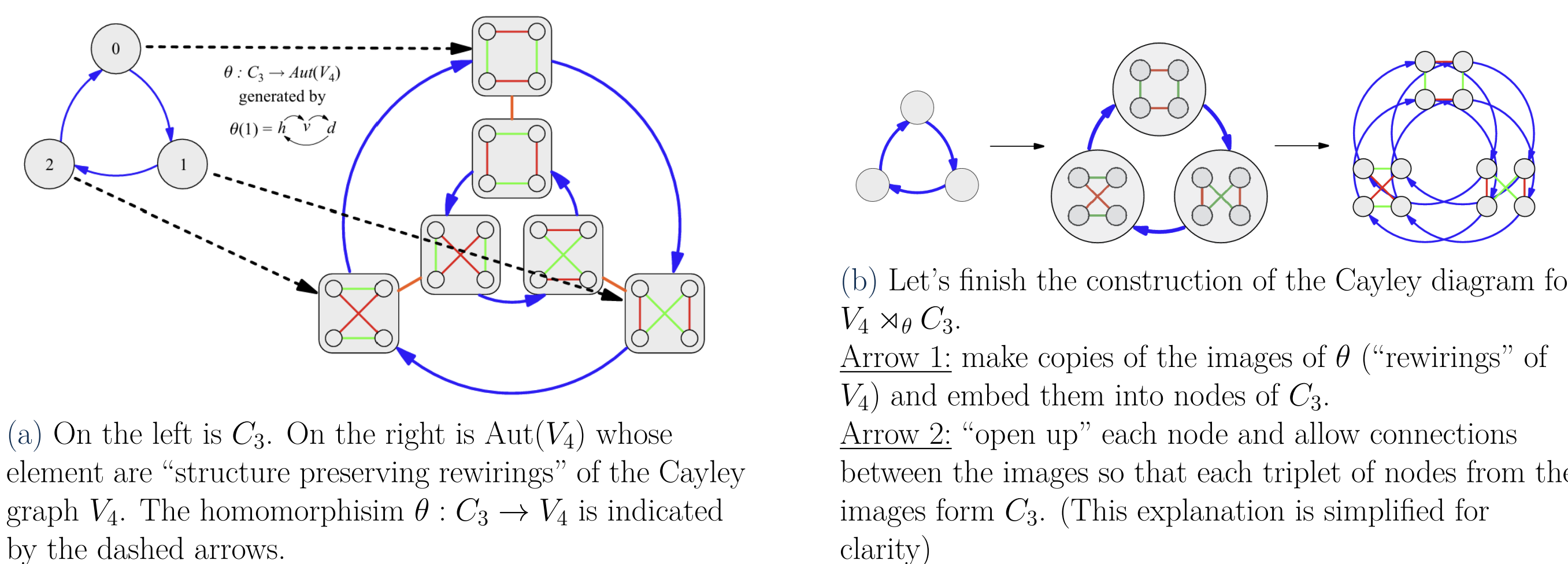


Figure 1. Constructing the Cayley Diagram of $V_4 \rtimes_{\theta} C_3$, with $\theta : C_3 \rightarrow \text{Aut}(V_4)$ a homomorphism.

Recognition Theorem of Semidirect Products

Theorem: Let G be a group with subgroups H and K such that

$$(1) H \triangleleft G$$

$$(2) G = HK$$

$$(3) H \cap K = \{e\}$$

Then $G \cong H \rtimes_{\phi} K$ where $\phi : K \rightarrow \text{Aut}(H)$ is conjugation: $\phi_k(h) = khk^{-1}$.

An Application: Classifying Groups of Order 30

Let G be a group of order 30. Sylow theorems shows that G has a normal subgroup $H \cong C_{15}$ and a subgroup $K \cong C_2$. $H \cap K = \{e\}$ by Lagrange's theorem and hence $G = HK$. The recognition theorem above tells us that $G \cong H \rtimes K$. To find all possible isomorphism types of G is to find all possible homomorphisms $\phi : K \rightarrow \text{Aut}(H)$. It turns out there're four of them, G is then isomorphic to one of:

$$G_{\phi_n} := C_{15} \rtimes_{\phi_n} C_2 \cong (C_5 \times C_3) \rtimes_{\phi_n} C_2, n \in \{1, 2, 3, 4\}.$$

$\text{Aut}(H) \cong \text{Aut}(C_{15}) \cong U_{15} \cong C_4 \times C_2$. Write $K = \langle k \rangle, H = \langle x \rangle \times \langle y \rangle$ ($C_5 = \langle x \rangle, C_3 = \langle y \rangle$). Keep in mind that homomorphisms are determined on the generators: we need to decide $\phi(k)$ is which element of $\text{Aut}(H)$. Since $\phi(k)$ has order 1 or 2, it could only be one of the following:

$$\begin{aligned}\phi_1(k) &: \{x \rightarrow x, y \rightarrow y\} \\ \phi_2(k) &: \{x \rightarrow x^{-1}, y \rightarrow y\} \\ \phi_3(k) &: \{x \rightarrow x, y \rightarrow y^{-1}\} \\ \phi_4(k) &: \{x \rightarrow x^{-1}, y \rightarrow y^{-1}\}\end{aligned}$$

Actually, these groups aren't exotic to us!

$$G_{\phi_1} \cong C_{30}, G_{\phi_2} \cong C_5 \times D_6, G_{\phi_3} \cong C_3 \times D_{10}, G_{\phi_4} \cong D_{30}$$

ϕ_1 is the trivial homomorphism, so by Example 1, $G_{\phi_1} = C_{15} \times C_2 \cong C_{30}$. G_{ϕ_2} has subgroups

$$\{(x, 0) : x \in C_5\} \cong C_5 \text{ and } \{((1, y), z) : y \in C_3, z \in C_2\} \cong C_3 \times C_2$$

where the second subgroup is isomorphic to D_6 by Example 2. Though all the groups above can be written as direct products, using semidirect products offers a complete classification avoiding complicated caseworks. [2].

References

- [1] N. C. Carter. The power of homomorphisms. In Visual Group Theory, chapter 8, pages 178–179. MAA, 1st edition, 2009.
- [2] R. M. F. David S.Dummit. Direct and semidirect products and abelian groups. In L. Rosatone, editor, Abstract Algebra, chapter 5, pages 178–182. John Wiley and Sons, 3rd edition, 2004.